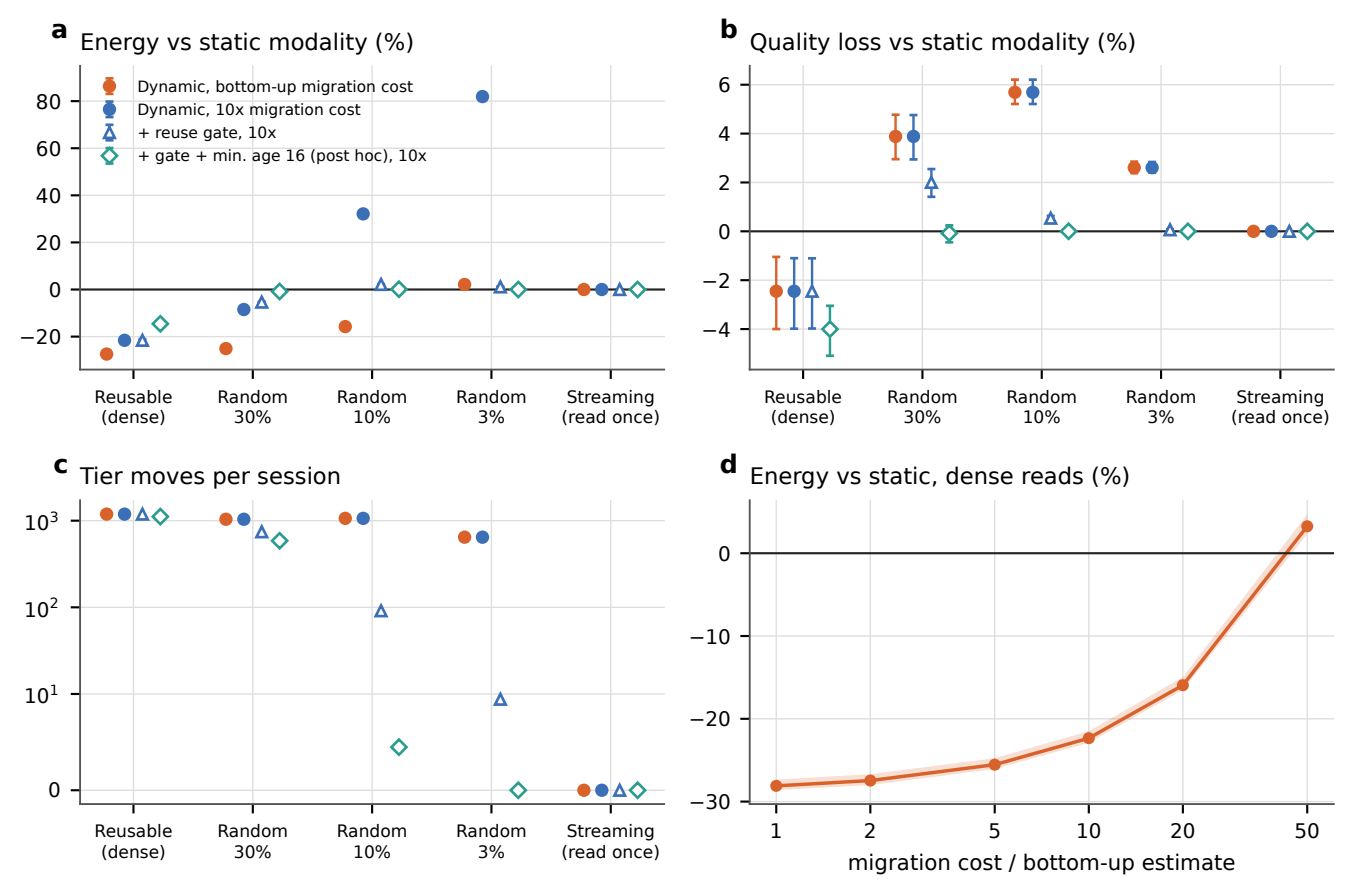
a
Energy vs static modality (%)
Dynamic, bottom-up migration cost
Dynamic, 10x migration cost
+ reuse gate, 10x
+ gate + min. age 16 (post hoc), 10x
80
60
40
20
0
−20
Reusable (dense)
Random 30%
Random 10%
Random 3%
Streaming (read once)
b
Quality loss vs static modality (%)
6
4
2
0
−2
−4
Reusable (dense)
Random 30%
Random 10%
Random 3%
Streaming (read once)
c
Tier moves per session
$10^3$
$10^2$
$10^1$
0
Reusable (dense)
Random 30%
Random 10%
Random 3%
Streaming (read once)
d
Energy vs static, dense reads (%)
0
−10
−20
−30
1
2
5
10
20
50
migration cost / bottom-up estimate